



SEA-BIRD
SCIENTIFIC

SBE37-SMP-ODO MicroCAT

Instrument Configuration

Instrument Serial Number: 37-16247
Instrument Firmware Version: 2.13.0
Zero Conductivity Frequency: 2819.01
Communications Format: RS232
Communications Settings: 9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

<i>Data Format</i>	<i>Measurement</i>	<i>Sensor Type</i>	<i>Serial Number</i>	<i>Rating</i>
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10746414	2000m(2000 dBar)
RS232	Oxygen	SBE 63	63-1788	7000m

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.



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SENSOR SERIAL NUMBER: 16247
CALIBRATION DATE: 03-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

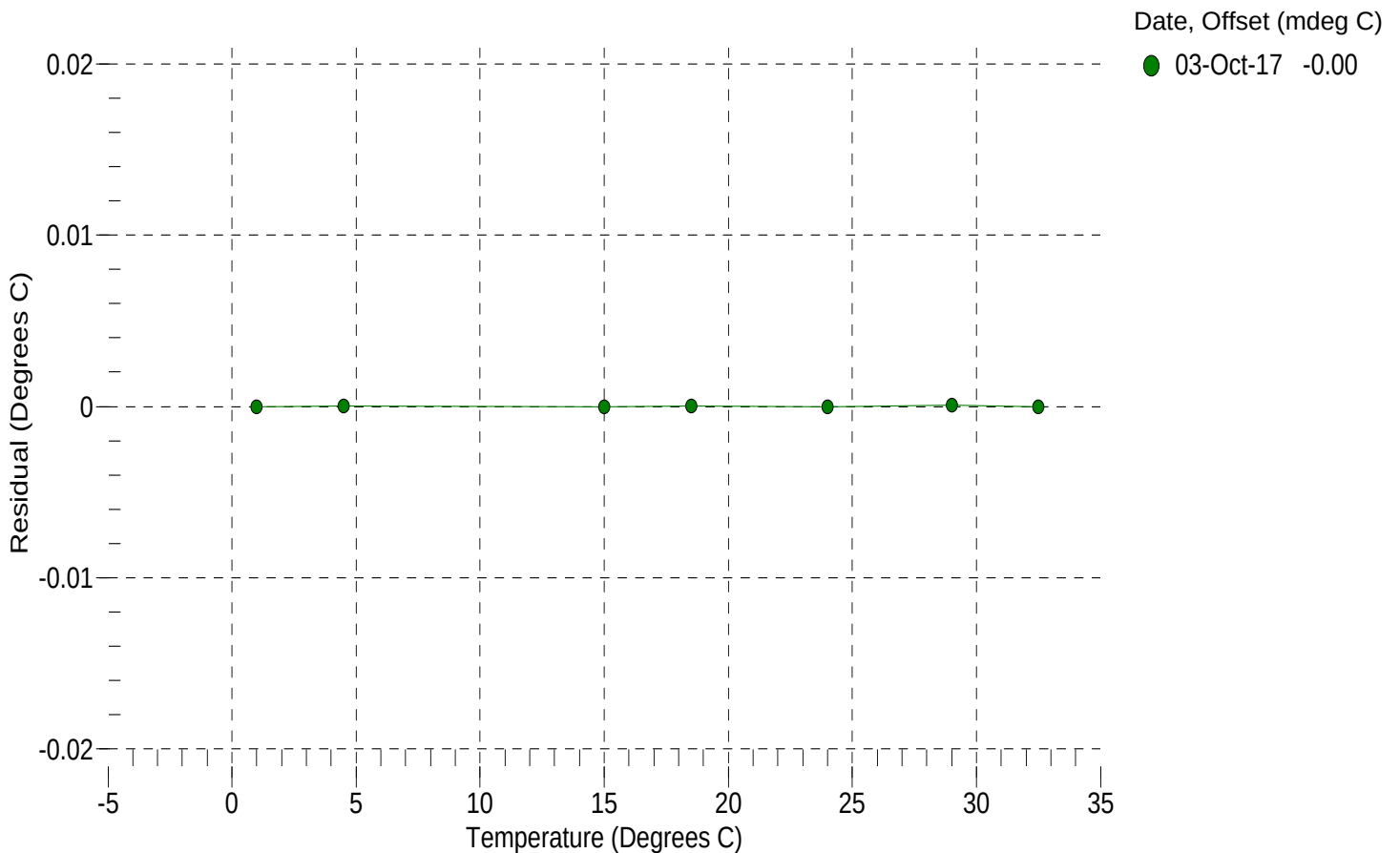
a0 = -1.187779e-004
a1 = 3.065352e-004
a2 = -4.363814e-006
a3 = 2.020303e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	569950.7	1.0000	-0.0000
4.5000	488141.1	4.5000	0.0000
15.0000	312683.3	15.0000	-0.0000
18.5000	271208.5	18.5000	0.0000
24.0000	218151.7	24.0000	-0.0000
29.0000	180065.8	29.0001	0.0001
32.5000	157956.6	32.5000	-0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature





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SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -9.680888e-001
h = 1.218970e-001
i = -8.695574e-005
j = 2.136662e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = -1.4779e-007

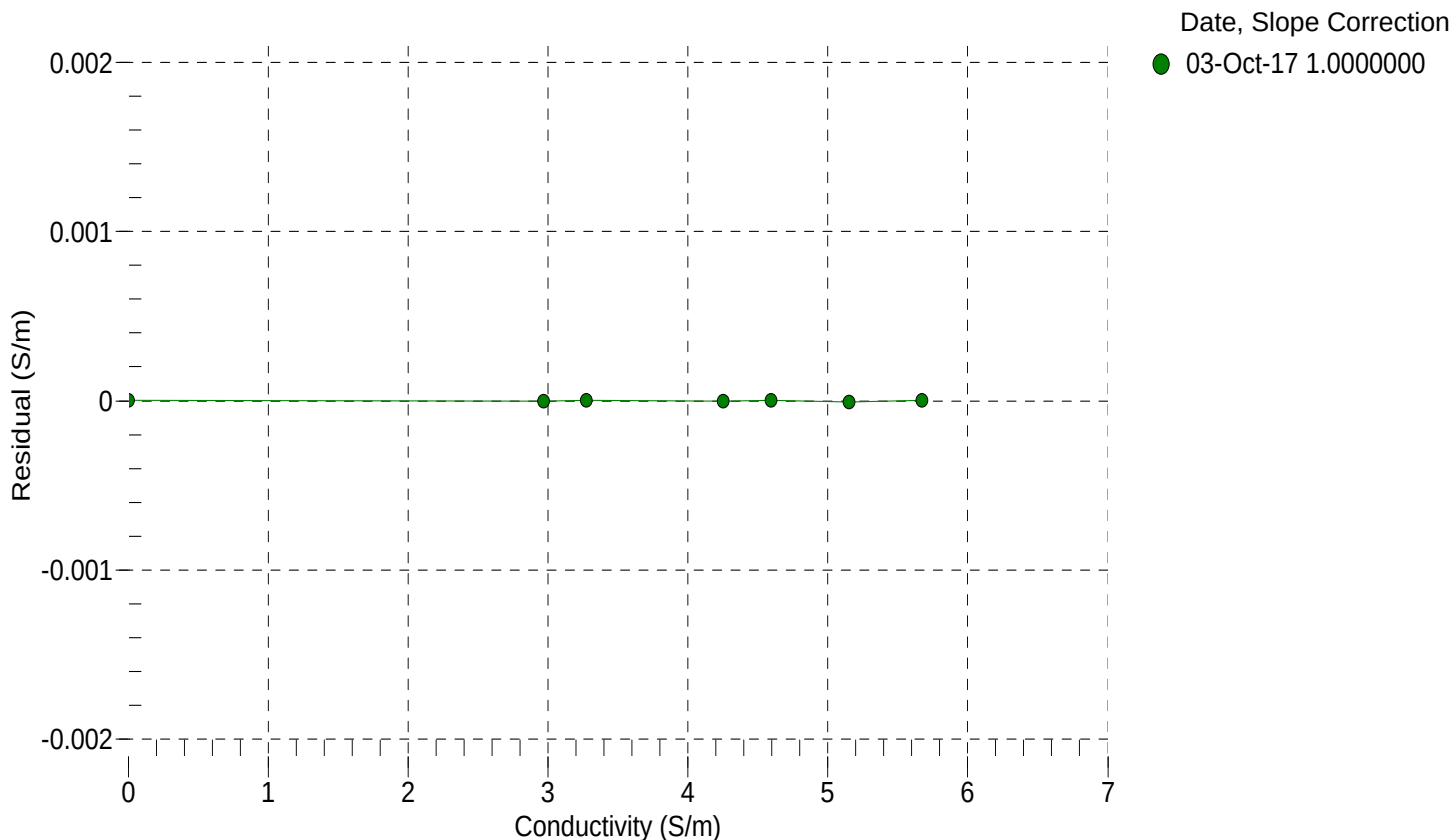
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2819.01	0.00000	0.00000
1.0000	34.6932	2.96644	5676.80	2.96644	-0.00000
4.5000	34.6732	3.27255	5892.69	3.27255	0.00000
15.0000	34.6306	4.25124	6534.43	4.25124	-0.00000
18.5000	34.6218	4.59534	6745.29	4.59535	0.00000
24.0000	34.6123	5.15162	7072.56	5.15162	-0.00001
29.0000	34.6073	5.67193	7365.20	5.67193	0.00000
32.5000	34.6050	6.04330	7566.86	6.04312	-0.00018

$f = \text{Instrument Output(Hz)} * \sqrt{1.0 + \text{WBOTC} * t} / 1000.0$

t = temperature (°C); p = pressure (decibars); $\delta = \text{CTcor}$; $\epsilon = \text{CPcor}$;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \epsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity





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SENSOR SERIAL NUMBER: 16247
CALIBRATION DATE: 29-Sep-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10746414

COEFFICIENTS:

PA0 =	1.981553e-001	PTCA0 =	5.241018e+005
PA1 =	9.089983e-003	PTCA1 =	-1.360471e+000
PA2 =	-1.511532e-010	PTCA2 =	1.109895e-001
PTEMPA0 =	-6.168716e+001	PTCB0 =	2.511536e+001
PTEMPA1 =	5.433889e-002	PTCB1 =	8.728180e-004
PTEMPA2 =	-6.863362e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
14.59	525715.3	1582.4	14.62	0.00	32.50	1773	525843.89
591.72	589316.0	1587.2	591.64	-0.00	29.00	1706	525831.37
1168.84	653058.5	1588.1	1168.73	-0.00	24.00	1610	525810.90
1746.00	716955.9	1589.3	1745.98	-0.00	18.50	1504	525787.32
2323.13	780985.2	1590.1	2323.20	0.00	15.00	1437	525774.93
2899.55	845041.8	1590.9	2899.41	-0.00	4.50	1238	525769.76
2323.07	780992.0	1590.9	2323.25	0.01	1.00	1171	525775.33
1746.03	716962.2	1590.3	1746.04	0.00	TEMPERATURE (°C) SPAN (mV)		
1168.85	653072.1	1590.1	1168.84	-0.00			
591.68	589322.7	1589.8	591.69	0.00			
14.59	525718.7	1590.6	14.63	0.00	-5.00		25.11
					35.10		25.15

y = thermistor output (counts)

t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y²

x = instrument output - PTCA0 - PTCA1 * t - PTCA2 * t²

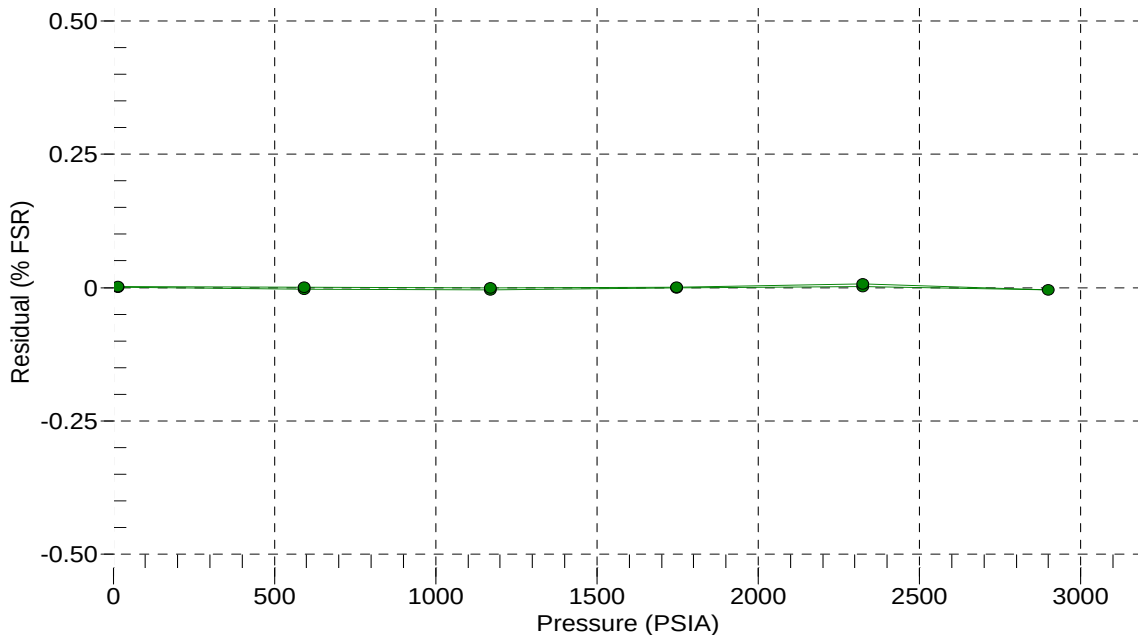
n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t²)

pressure (PSIA) = PA0 + PA1 * n + PA2 * n²

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range

Date, Offset (%FSR)

● 29-Sep-17 -0.00





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Pressure Test Certificate

Test Date: **2017-09-28**

Description: **SBE-37 Microcat**

Sensor Information:

Model Number: **SBE-37**

Serial Number: **16247**

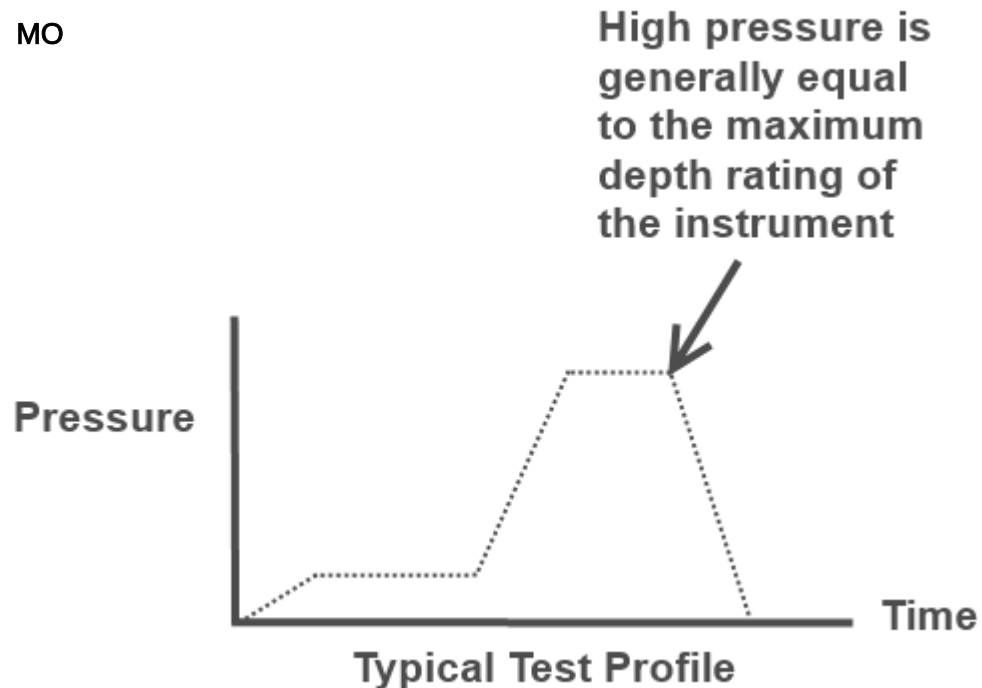
Pressure Test Protocol:

Low Pressure Test: **40** PSI Held For: **15** Minutes

High Pressure Test: **2900** PSI Held For: **15** Minutes

Passed Test: **True**

Tested By: **MO**





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SENSOR SERIAL NUMBER: 1788
CALIBRATION DATE: 08-Sep-17

SBE 63 OXYGEN TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

TA0 = 6.993372e-004 TA2 = 6.227171e-007
TA1 = 2.514323e-004 TA3 = 1.044262e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT(V)	INST TEMP (° C)	RESIDUAL (° C)
1.9999	1.12305	1.9997	-0.00016
1.9999	1.12305	1.9997	-0.00016
2.0000	1.12303	2.0004	0.00036
2.0001	1.12304	2.0000	-0.00005
5.9999	0.99867	5.9999	-0.00002
5.9999	0.99867	5.9999	-0.00002
6.0000	0.99867	5.9999	-0.00012
6.0000	0.99866	6.0002	0.00022
11.9999	0.83250	11.9998	-0.00005
11.9999	0.83250	11.9998	-0.00005
12.0000	0.83250	11.9998	-0.00015
12.0001	0.83249	12.0002	0.00014
19.9999	0.64835	19.9998	-0.00010
20.0000	0.64834	20.0003	0.00028
20.0000	0.64835	19.9998	-0.00020
20.0001	0.64834	20.0003	0.00018
26.0000	0.53598	26.0001	0.00014
26.0000	0.53599	25.9995	-0.00045
26.0000	0.53598	26.0001	0.00014
26.0001	0.53598	26.0001	0.00004
30.0000	0.47193	30.0001	0.00011
30.0001	0.47193	30.0001	0.00001
30.0001	0.47193	30.0001	0.00001
30.0002	0.47193	30.0001	-0.00009

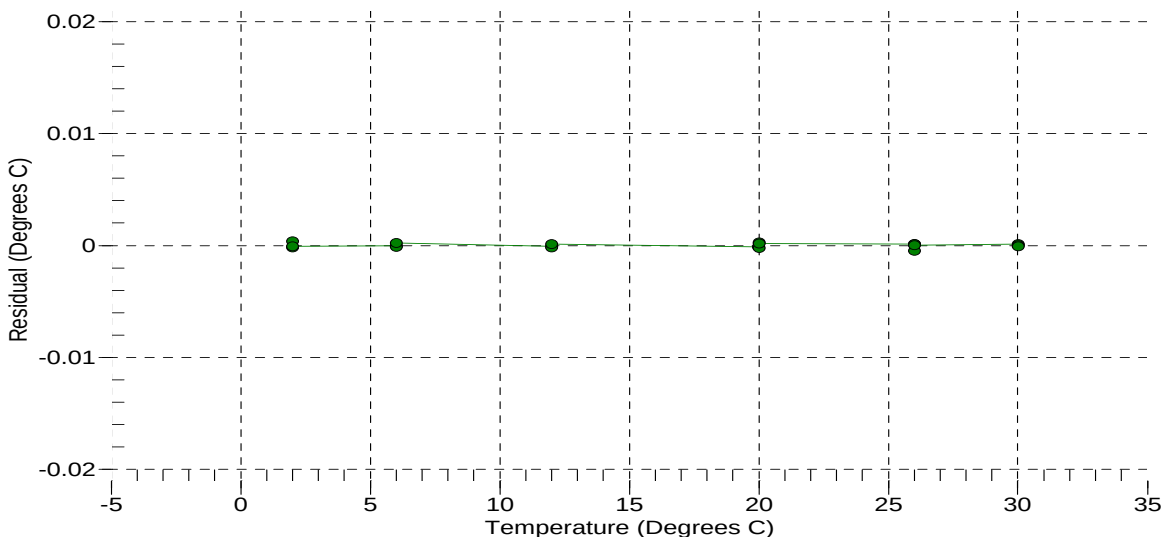
V = Instrument Output (Volts)

$L = \ln(100000 * V / (3.3 - V))$

Temperature ITS-90 (°C) = $1 / (TA0 + (TA1 * L) + (TA2 * L^2) + (TA3 * L^3)) - 273.15$

Residual (°C) = instrument temperature - bath temperature

Date, Offset (mdeg C)
● 08-Sep-17 -0.00





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SBE 63 OXYGEN CALIBRATION DATA

COEFFICIENTS:

A0 = 1.0513e+000 B0 = -2.4700e-001 C0 = 9.5700e-002 E = 1.1000e-002
A1 = -1.5000e-003 B1 = 1.6856e+000 C1 = 4.0724e-003
A2 = 4.5629e-001 C2 = 5.5191e-005

BATH OXYGEN (ml/l)	BATH TEMPERATURE (° C)	BATH SALINITY (PSU)	INSTRUMENT OUTPUT (µsec)	INSTRUMENT OXYGEN (ml/l)	RESIDUAL (ml/l)
0.715	30.00	0.00	31.18	0.710	-0.005
0.748	26.00	0.00	31.80	0.743	-0.005
0.796	20.00	0.00	32.80	0.793	-0.003
0.883	12.00	0.00	34.16	0.881	-0.002
0.988	6.00	0.00	35.07	0.991	0.003
1.083	2.00	0.00	35.64	1.092	0.009
2.219	30.00	0.00	22.80	2.220	0.001
2.341	26.00	0.00	23.37	2.344	0.003
2.490	20.00	0.00	24.51	2.493	0.003
3.001	12.00	0.00	25.43	2.995	-0.006
3.409	6.00	0.00	26.40	3.401	-0.008
3.691	30.00	0.00	18.85	3.696	0.005
3.773	2.00	0.00	27.01	3.766	-0.008
3.936	26.00	0.00	19.30	3.941	0.005
4.349	20.00	0.00	20.05	4.361	0.012
5.104	12.00	0.00	21.10	5.104	0.001
5.294	30.00	0.00	16.30	5.282	-0.012
5.646	26.00	0.00	16.69	5.641	-0.005
5.845	6.00	0.00	21.94	5.846	0.001
6.280	20.00	0.00	17.31	6.288	0.008
6.471	2.00	0.00	22.52	6.474	0.003
7.346	12.00	0.00	18.27	7.341	-0.005
8.397	6.00	0.00	19.06	8.393	-0.004
8.922	2.00	0.00	19.91	8.926	0.004

T = temperature (°C) , P = pressure (dbar), U = Instrument output (µsec)

S_{corr} (salinity correction function) = 1.0 for calibration in DI water

See the user manual for more information on S_{corr} calculation

$V = U / 39.457071$

Oxygen (ml/l) = $\{((A0 + A1 \cdot T + A2 \cdot V^2) / (B0 + B1 \cdot V) - 1.0) / (C0 + C1 \cdot T + C2 \cdot T^2)\} \cdot S_{corr} \cdot \exp(E \cdot P / (T + 273.15))$

Residual (ml/l) = instrument oxygen - bath oxygen

Date, Slope Correction

● 08-Sep-17 1.0000

